

312 PROBLEM SET 1

QUESTIONS FOR DISCUSSION

1. Define and give the dimensions of thermal conductivity k , thermal diffusivity α , heat capacity $\hat{C}_p$, heat flux q , and combined energy flux e . For the dimensions use m = mass, l = length, T = temperature, and t = time.
2. Compare the orders of magnitude of the thermal conductivities of gases, liquids, and solids.
3. In what way are Newton's law of viscosity and Fourier's law of heat conduction similar? Dissimilar?
8. Is the relation $C_p - C_v = R$ true only for ideal gases, or is it also true for liquids? If it is not true for liquids, what formula should be used?
10. What is $[\pi \cdot v] = p v + [\tau \cdot v]$ for Poiseuille flow?

①

Thermal Conductivity

$$k [=] \frac{m \cdot l}{T \cdot t^3}$$

$$\left(q_x = -k \frac{dT}{dx} \right)$$

↓

How easily heat flows through a material through conduction.

Thermal Diffusivity

$$\alpha = \frac{k}{\rho C_p} [=] \frac{l^2}{t}$$

$$\left(\frac{\text{area}}{\text{time}} \right)$$

↓

How fast temperature changes spread through a material

Heat Capacity

$$\hat{C}_p [=] \frac{\text{Energy}}{m \cdot T} = \frac{m l^2 t^{-2}}{m T} [=] \frac{l^2}{t^2 \cdot T}$$

↓

How much energy is needed to raise the temperature of a material

Heat flux

$$q = -k \frac{dT}{dx} [=] \frac{\text{Energy}}{\text{time} \cdot \text{area}} [=] \frac{m \cdot l}{T \cdot t^3} \cdot \frac{T}{l} [=] \frac{\text{mass}}{t^3}$$

Heat transfer per unit time per unit area

combined energy flux

↓

conduction + convection

$$e = q + \rho C_p \cdot v \cdot T$$

$$e [=] \text{ g } [=] \frac{\text{m}}{\text{s}^3}$$

(2)

$$\kappa_{\text{gas}} \ll \kappa_{\text{liquid}} \ll \kappa_{\text{solid}}$$
$$10^{-2} \ll 10^{-1} \ll 10^0 - 10^2$$

(3)

Newton's Law: $\tau_x = -\eta \frac{dv}{dx}$

Fourier's Law: $q_x = -\kappa \frac{dT}{dx}$

Similarities:

- both say flux is proportional to a driving gradient
- both fluxes flow from high to low

Differences:

- Newton's law: Describes molecular momentum transport $\propto$ velocity gradient, viscosity
- Fourier's Law Describes heat transport $\propto$ temp gradient, thermal conductivity

(8)

$$C_p - C_v = R \quad \text{only for ideal gases}$$

$$C_p - C_v = \frac{T \alpha^2}{\beta \rho}$$

α : thermal expansion coefficient
 β : isothermal compressibility

for ideal gases:

$$C_p - C_v = \frac{T \cdot \frac{1}{T^2}}{\frac{1}{P} \cdot \frac{1}{P}} \quad \left\{ \begin{array}{l} \alpha = \frac{1}{T} \\ \beta = \frac{1}{P} \end{array} \right.$$

$$C_p - C_v = \frac{\frac{1}{T}}{\frac{1}{V} \cdot \frac{1}{P}} = \frac{PV}{T} = R \quad \left(\begin{array}{l} PV = RT \\ P = \frac{PV}{T} \end{array} \right)$$

(1D)

for poiseuille flow:

velocity only in z direction $V \Rightarrow v_z(r)$

$$\text{and } \tau_{rz} = -\mu \frac{dv_z}{dr} \quad , \quad \tau_{rz} \cdot v_z(r)$$

$$[\pi \cdot v] = p \cdot v_z(r) + \left(-\mu \frac{dv_z}{dr} \cdot v_z(r) \right)$$